

Unit IV

4

Operating System

4.1 : Introduction

Q.1 What is operating system ? Draw and explain layered approach with advantages and disadvantages. [SPPU : May-19, End Sem, Marks 6]

Ans. : • Operating system is a program that controls the execution of application programs. It is an interface between applications and hardware.

• OS provides different types of view. For user, it is abstract view because it provides features which are important for users. OS is intermediary between user and the computer system.

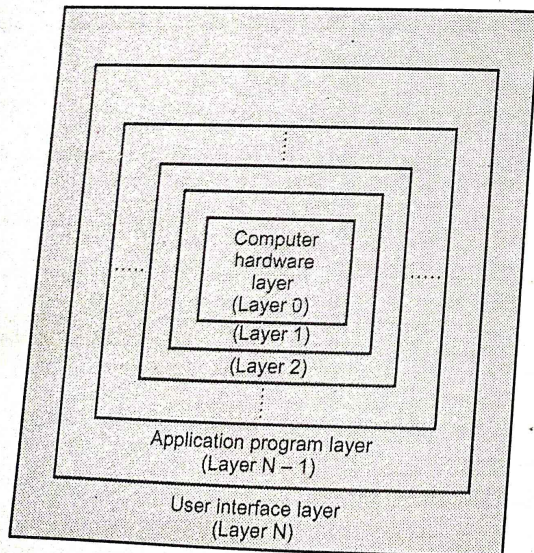


Fig. Q.1.1 Operating system layers

- Operating system is divided into number of layers. Each layer boundary is properly defined. Bottom layer is called layer 0 and top most layer is called layer N. Layer N provides user interface.
- A function of layer is also fixed. Fig. Q.1.1 shows the operating system layer. Each layer consists of data structure and a set of routines. Layer provides services to upper and lower layers. (See Fig. Q.1.1 on previous page.)
- First layer contains only basic hardware to implement function. So only first layer is debugged for checking the whole system. If error is not found then the system will work properly. If error is encountered while debugging second layer, then error is related to the second layer only.
- In this operating system is designed and implemented. It simplifies the design task of the operating system.
- Each layer uses system calls for performing their operation. Layer adds overhead to the system call.

Advantages of the layered approach to system design.

1. Modularity is the advantage of the layered system.
2. The system is easier to debug and modify because changes affect only limited sections of the system rather than touching all sections of the operating system.
3. Information is kept only where it is needed and is accessible only within a defined and restricted area, so any bugs affecting that data must be limited to a specific module or layer.

Q.2 Why operating system is needed ? Justify "Operating system is the control program of the system."

Ans. : • Operating system is needed because it allows a program to run on multiple different types of hardware. An operating system provides a layer between the program and hardware to enable a program to use a standard interface no matter what hardware is used. The operating system translates the software calls to the hardware calls for that particular piece of hardware.

- The operating system is responsible for managing all resources. A portion of the OS is in main memory. This portion of the OS is called kernel.
- User program and data is also stored in remaining parts of the memory. Allocation of main memory is controlled by operating system with the help of memory management hardware.

- I/O device is controlled by OS and it decides when an I/O device can be used by program in execution. Processor is one type of resource and OS control the execution of user program on the processor.
- Modern OS allows multiple programs to run at the same time. If multiple users are using computer then there is need of managing and protecting the memory, I/O devices and other resources.

Q.3 What are the major functions of operating system ?

Ans. : The major functions of an operating system are :

1. Efficient use of a computer system
2. User convenience
3. Ability to evolve

1. **Efficient use :**
 - For efficient use of resources, it must be monitored by operating system. Proper scheduling of resources is also required.
 - Computer contains different types of resources like CPU, memory and I/O device etc. Proper monitoring is required on these resources to avoid the overhead. As per the resource, scheduling is required.
2. **User convenience :**
 - User convenience is affected by computing environment of the computer system. The computing environment is comprised of computer system, its interfaces with other systems and nature of computations performed by its users.
 - Computer architecture and use change the computing environment of the system.
3. **Ability to evolve :**
 - An OS should be constructed in such a way as to permit the effective development, testing and introduction of new system functions without at the same time interfering with service.

Q.4 Explain operating system as a resource manager.

Ans. : • A computer is a set of resources. These resource provides various functions to the user. Functions like data movement, storing of data and program, operation on data are control by an operating system.

- Fig. Q.4.1 (See Fig. Q.4.1 on next page.) shows OS as a resource manager.
- The operating system is responsible for managing the all resources. A portion of the OS is in main memory. This portion of the OS is called kernel.
- User program and data is also stored in remaining parts of the memory. Allocation of main memory is controlled by operating system with the help of memory management hardware.

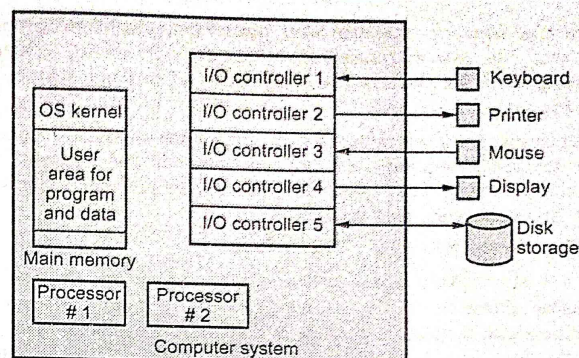


Fig. Q.4.1 OS as a resource manager

- I/O device is controlled by OS and it decides when an I/O device can be used by program in execution. Processor is one type of resource and OS control the execution of user program on the processor.
- Modern OS allows multiple programs to run at the same time. If multiple users are using computer then there is need of managing and protecting the memory, I/O devices and other resources.
- Resource management includes sharing resources in different ways. Time and space are the two concept for resource sharing.
 1. **Time :** Time slot is allocated to each program first one gets to use the resource then another and so on.
 2. **Space :** Consider the example of main memory. Main memory is normally divided up among several running programs, so each one can be resident at the same time.

4.2 : Evolution of OS

Q.5 Explain following term clearly : Multiprogramming.

Ans. : • When two or more programs are in memory at the same time, sharing the processor is referred to the multiprogramming operating system. Multiprogramming assumes a single processor that is being shared. It increases CPU utilization by organizing jobs so that the CPU always has one to execute.

- Resource management is the main aim of multiprogramming operating system. File system, command processor, I/O control system and transient area are the essential components of a single user operating system.
- Multiprogramming operating system divides the transient area to store the multiple programs and provides resource management to the operating system. The concurrent execution of programs improves the utilization of system resources.

Q.6 Explain the following : Multitasking.

Ans. : • Multitasking is also called as time sharing operating system. It is logical extension of the multiprogramming operating systems. User interaction with program is possible in time sharing operating system.

- During execution of the program, user interacts directly with the program, supplying information to the program.
- Multi-tasking means that the computer can work with more than one program at a time. For example, user could be working with information from one database on the screen analyzing data, while the computer is sorting information from another database, while a excel sheet is performing calculations on a separate worksheet.
- Many users share the computer system simultaneously in time sharing operating system. Time sharing system uses multiprogramming and CPU scheduling. Each user has at least one separate program in memory.
- Concept of virtual machine is used in time sharing system. It creates virtual machine one per user. User interaction with system by using virtual machine. User enters the command for virtual machine and result will received back to user.
- Time sharing system is more complex than multiprogramming operating system. It also takes help of file system. File system is stored on the disk so disk management is also required.

Q.7 Write down advantages of batch processing system.

Ans. :

1. Move much of the work of the operator to the computer.
2. Increased performance since it was possible for job to start as soon as the previous job finished.
3. Batch system often provides simple forms of file management.

4. Batch systems do not require any time critical device management.
5. Scheduling is also simple in batch system. Jobs are processed in the order of submission i.e. first come first served fashion.

Q.8 What do you mean spooling? Explain in detail.

Ans. : • When a job completes execution, its memory is released and the output for the job gets copied into an outputs pool for later printing.

- Spooling an acronym for simultaneous peripheral operation on line. Spooling uses the disk as a large buffer for outputting data to printers and other devices. It can also be used for input, but is generally used for output.
- Its main use is to prevent two users from alternating printing lines to the line printer on the same page, getting their output completely mixed together. It also helps in reducing idle time and overlapped I/O and CPU.
- Spooling refers to putting jobs in a buffer, a special area in memory or on a disk where a device can access them when it is ready.
- Spooling is useful because device access data at different rates. The buffer provides a waiting station where data can rest while the slower device catches up.
- Computer can perform I/O in parallel with computation, it becomes possible to have the computer read a deck of cards to a tape, drum or disk and to write out to a tape printer while it was computing. This process is called spooling.
- The most common spooling application is print spooling. Spooling batch system were the first and are the simplest of the multiprogramming systems.
- In spooling, I/O of one job is overlapped with the computation of another job. For example, a spooler at a time may read input of one job, and at the same time, it may also print the output of another job.
- Spooling can also process data at the remote sites. The spooler only has to notify when a process gets completed at the remote site so that spooler can spool next process to the remote side device.
- Spooling increases the performance of the system by increasing the working rate of the devices. It naturally leads to multiprogramming.

Advantages of spooling

- The spooling operation uses a disk as a very large buffer.
- Spooling is however capable of overlapping I/O operation for one job with processor operations for another job.

Q.9 Explain features of RTOS and time sharing operating system.

Ans. : Features of Real Time Operating System :

1. Real time system fails if it does not give result within the time limits.
2. Real time task cannot keep waiting for longer time without allocating kernel.
3. Real time operating system uses priority scheduling algorithm.

Features of Time Sharing Operating System :

1. User interaction with program is possible in time sharing operating system.
2. Time sharing system uses medium term scheduling such as round robin for the foreground.
3. In time sharing system, each user is given a time slice for executing his/her job in round robin fashion.

Q.10 What is real time OS ? Explain its types with suitable example.

[SPPU : May-19, End Sem, Marks 4]

Ans. : • Time constraints is the key parameter in real time systems. It controls autonomous system such as robots, satellites, air traffic control and hydroelectric dams.

- When user gives an input to the system, it must process within the time limit and result is sent back. Real time system fails if it does not give result within the time limits.
- A real-time system is any information processing system which has to respond to externally generated input stimuli within a finite and specified period.
- Real time systems are of two types : Hard and soft.

1. Hard Real Time Systems :

- A hard real-time system is one where the response time is specified as an absolute value. This time is normally dictated by the environment.

- A system is called a hard real-time if tasks always must finish execution before their deadlines or if message always can be delivered within a specified time interval.
- Hard real-time is often associated with safety critical applications. A failure (e.g. missing a deadline) in a safety-critical application can lead to loss of human life or severe economical damage.

2. Soft Real Time System :

- A soft real-time system is one where the response time is normally specified as an average value. This time is normally dictated by the business or market.
- A single computation arriving late is not significant to the operation of the system, though many late arrivals might be.
- Soft real time means that only the precedence and sequence for the task-operations are defined, interrupt latencies and context switching latencies are small but there can be few deviations between expected latencies of the tasks and observed time constraints and a few deadline misses are accepted.

Q.11 What is distributed system ? Explain its advantages.

Ans. : • Definition : A distributed system is a collection of autonomous hosts that are connected through a computer network.

- A distributed system is a collection of independent computers that appears to its users as a single coherent system. Each host executes components and operates a distribution middleware.
- Middleware enables the components to coordinate their activities. Users perceive the system as a single, integrated computing facility.
- A distributed computer system consists of multiple software components that are on multiple computers, but run as a single system. The computers that are in a distributed system can be physically close together and connected by a local network, or they can be geographically distant and connected by a wide area network.
- A distributed system can consist of any number of possible configurations, such as mainframes, personal computers, workstations, minicomputers and so on.
- Distributed operating systems depend on networking for their operation. Distributed OS runs on and controls the resources of multiple machines. It provides resource sharing across the boundaries of a single computer system. It looks to users like a single machine OS.

- Distributing OS owns the whole network and makes it look like a virtual uni-processor or may be a virtual multiprocessor.
- Definition : A distributed operating system is one that looks to its users like an ordinary operating system but runs on multiple, independent CPU.
- Distributed systems depend on networking for their functionality. Fig. Q.11.1 shows the distributed system.

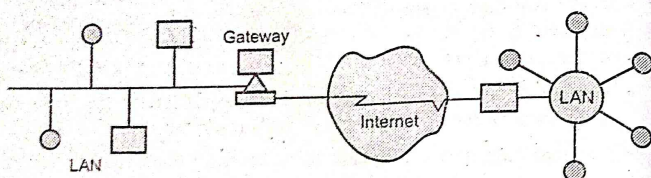


Fig. Q.11.1 Distributed system

Advantages of distributed OS :

1. **Resource sharing :** Sharing of software resources such as software libraries, database and hardware resources such as hard disks, printers and CDROM can also be done in a very effective way among all the computers and the users.
2. **Higher reliability :** Reliability refers to the degree of tolerance against errors and component failures. Availability is one of the important aspects of reliability. Availability refers to the fraction of time for which a system is available for use.
3. **Better price performance ratio :** Reduction in the price of microprocessor and increasing computing power gives good price-performance ratio.
4. **Shorter responses times and higher throughput.**
5. **Incremental growth :** To extend power and functionality of a system by simply adding additional resources to the system.

Q.12 What is Real Time Operating System ? What is the difference between Hard real time and Soft real time operating system ?

Ans. : Real time system is one that must react to inputs and responds to them quickly. A real time system has well defined, fixed time constraints.

Difference Between Hard and Soft Real Time System

Sr. No	Hard real time	Soft real time
1.	Hard response time is required.	Soft response time is required.
2.	Data integrity is short term.	Data integrity is long term.
3.	Size of data file is small or medium.	Size of data file is large.
4.	Peak load performance is predictable.	Peak load performance is degraded.
5.	Hard real-time systems have little laxity and generally provide full deadline compliance.	Soft real-time systems are more flexible. They have greater laxity and can tolerate certain amounts of deadline misses.
6.	Safety critical systems are typically hard real time systems.	Linux and many OS provide soft real time system.

4.3 : Operating System Services

Q.13 What are the different services provided by the operating system ?

Ans. : 1. **Program execution :** Before executing the program, it is loaded into the memory by operating system. Once program loads into memory, its start execution.

2. **Input : output operation :** Program is combination of input and output statement. While executing the program, it requires I/O device.

3. **Error detection :** Error is related to the memory, CPU, I/O device and in the user program. Memory is full, stack overflow, file not found, directory not exist, printer is not ready, attempt to access illegal memory are the example of error detection.

4. **File and directory operation :** User wants to read and writes the file and directory.

5. **Communication :** Communication may be inter-process communication and any other type of communication. Data transfer is one type of communication.

4.4 : Process Management : Process, Process States : 5 and 7 State Model, Process Control Block

Q.14 Explain the following terms clearly : Process.

Ans. : • Process is an active entity that requires a set of resources, including a processor, program counter, registers to perform its function. Multiple processes may be associated with one program.

• Process means a program in execution. Process execution must progress in sequential order. It also called task. Task is a single instance of an executable program.

• Each process has its own address space. Address space is divided into regions :

1. **Text region :** It stores the code that the processor executes.

2. **Data region :** It stores variables and dynamically allocated memory that the process uses during execution.

3. **Stack region :** It stores instructions and local variables for active procedure calls.

• Process is a dynamic entity that executes a program on a particular set of data using resources allocated by the operating system.

• A process can run to completion only when all requested hardware and software resources have been allocated to the process.

Q.15 Explain process control block in detail.

[SPPU : May-19, End Sem, Marks 6]

OR Explain process control block with suitable diagram.

[SPPU : May-18, End Sem, Marks 6]

Ans. : Process control block : When a process is created by operating system, it allocates a PCB for it. OS initializes PCB and puts PCB on the correct queue. Fig. Q.15.1 shows PCB.

Following information is stored in process control block.

1. **Process identification :** Each process is uniquely identified by the user's identification and a pointer connecting it to its descriptor.
2. **Priority number :** Operating system allocates the priority number to each process. According to the priority number it allocates the resources.

3. **Program counter :** The PC indicates the address of the next instruction to be executed for this current process.

4. **Memory allocation :** It contains the value of the base registers, limit registers and the page tables depending on the memory system used by the operating system.

5. **I/O status information :** It maintains information about the open files, list of I/O devices allocated to the process etc.

6. **List of open files :** Process uses number of files for operation.

7. **Process state :** Process may be in any one of the state : New, ready, running, and waiting, terminate.

Process identification
Priority number
Program counter
Memory allocation
I/O status information
List of open files
Accounting information
Number of registers
Process state

Fig. Q.15.1 Process control block

Q.16 Draw and explain process state transition diagram.

[SPPU : May-18, End Sem, Marks 6]

Ans. :

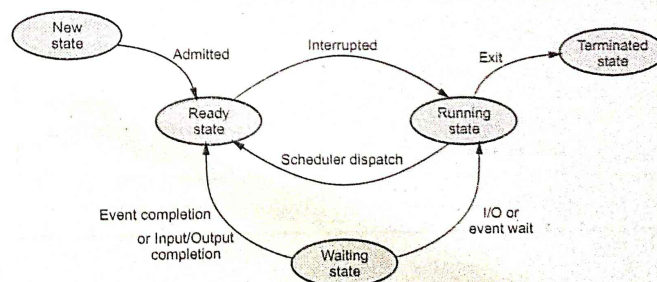


Fig. Q.16.1 Process state diagram

- Fig. Q.16.1 shows a process state diagram. A state diagram is composed of a set of states and transitions between states.
- State diagram is used by process manager to determine the type of service to provide to the process.
- The process states are as follows :
1. New 2. Ready 3. Running 4. Waiting 5. Terminated
- **New** : Operating system creates new process by using fork() system call. These process are newly created process and resources are not allocated.
- **Ready** : The process is competing for the CPU. Process reaches to the head of the list (queue).
- **Running** : The process that is currently being executed. Operating system allocates all the hardware and software resources to the process for execution.
- **Waiting** : A process is waiting until some event occurs such as the completion of an input-output operation.
- **Terminated** : A process completes its operations and releases all resources.

Q.17 Differentiate between the following : Process and program.

Ans. : Difference between process and program

Sr. No.	Process	Program
1.	Process is active entity.	Program is passive entity.
2.	Process is a sequence of instruction executions.	Program contains the instructions.
3.	Process exists in a limited span of time.	A program exists at single place and continues to exist.
4.	Process is a dynamic entity.	Program is a static entity.

4.5 : Threads

Q.18 What is process ? What is thread ? List down benefits of using thread.

Ans. : Process : Refer Q.14. [SPPU : May-18, End Sem, Marks 6]

Thread :

- Thread is a dispatchable unit of work. It consists of thread ID, program counter, stack and register set. Thread is also called a Light Weight Process (LWP). Because they take fewer resources than a process. A thread is easy to create and destroy.
- It shares many attributes of a process. Threads are scheduled on a processor. Each thread can execute a set of instructions independent of other threads and processes. Fig. Q.18.1 shows a thread.
- Every program has at least one thread. Programs without multithreading executes sequentially. That is, after executing one instruction the next instruction in sequence is executed.

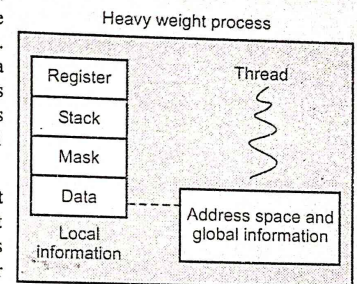


Fig. Q.18.1 Thread

- Thread is a basic processing unit to which an operating system allocates processor time and more than one thread can be executing code inside a process.
- Each thread belongs to exactly one process and no thread can exist outside a process. Each thread represents a separate flow of control.
- Threads are of two types : User level thread and Kernel level thread.

Benefit of using thread :

1. Context switching time is minimized.
2. Thread support for efficient communication.
3. Resource sharing is possible using threads.

4. A thread provides concurrency within a process.
 5. It is more economical to create and context switch threads.

Q.19 How thread is different from process?

Ans. :

Sr. No.	Thread	Process
1.	Thread is also called lightweight process.	Process is also called heavyweight process.
2.	Operating system is not required for thread switching.	Operating system interface is required for process switching.
3.	One thread can read, write or even completely clean another threads stack.	Each process operates independently of the other process.
4.	All threads can share same set of open files and child processes.	In multiple processing, each process executes the same code but has its own memory and file resources.
5.	If one thread is blocked and waiting then second thread in the same task can run.	If one server process is blocked then other server process cannot execute until the first process is unblocked.
6.	Uses fewer resources.	Uses more resources.

Q.20 Explain thread lifecycle.

Ans. : • Fig. Q.20.1 shows a thread lifecycle. A thread has one of the following states. (See Fig. Q.20.1 on next page.)

- 1. New :** Thread is just created.
- 2. Ready :** Thread's start () method invoked and now it is executing. OS put thread into ready queue.
- 3. Running :** Highest priority ready thread enters the running state. Thread is assigned a processor and now is running.
- 4. Blocked :** This is the state when a thread is waiting for a lock to access an object.

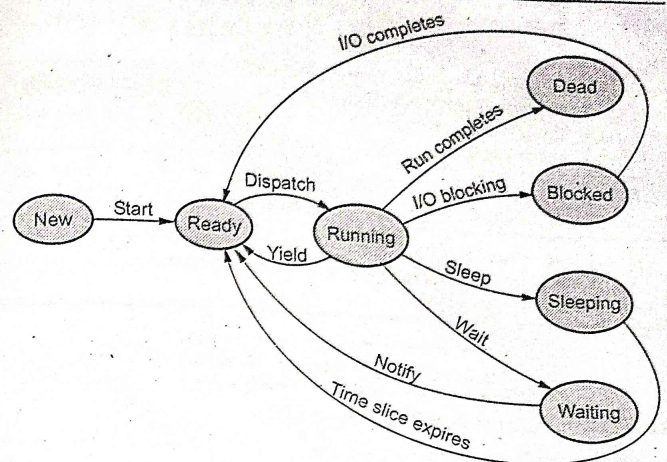


Fig. Q.20.1 Thread lifecycle

- 5. Waiting :** Here thread is waiting indefinitely for another thread to perform an action.
- 6. Sleeping :** Thread sleep for a specified time of period. When sleeping time expires, it enters to ready state. CPU is not used by sleeping thread.
- 7. Dead :** When thread completes task or operation.

Q.21 What is difference between user level thread and kernel level thread ?

Ans. :

Sr. No.	User level threads	Kernel level threads
1.	User level threads are faster to create and manage.	Kernel level threads are slower to create and manage.
2.	Implemented by a thread library at the user level.	Operating system support directly to Kernel threads.
3.	User level thread can run on any operating system.	Kernel level threads are specific to the operating system.

4.	Support provided at the user level called user level thread.	Support may be provided by Kernel is called Kernel level threads.
5.	Multithread application cannot take advantage of multiprocessing.	Kernel routines themselves can be multithreaded.
6.	Example : POSIX P threads and Mach C-threads	Example : Windows 95/98/NT, Sun Solaris and Digital UNIX
7.	User level threads are also called many to one mapping thread.	Kernel level thread support one to one thread mapping.

4.6 : Multithreading Model

Q.22 Explain various multithreading model.

Ans. : • Operating system uses user level thread and kernel level thread. User level threads are managed without kernel support. Operating system support and manage the kernel level threads.

• Modern operating system support kernel level threads. Kernel performs multiple simultaneous tasks in operating system. In most of the application, user threads are mapped with kernel level threads.

• Different methods of mapping is used in operating system are as follows :

1. One to one 2. Many to one 3. Many to many

1. One to one

- In this model, one user level thread maps with one kernel level thread. If there are three user threads then system creates three separate kernel thread. This model provides more concurrency because of separate thread. Fig. Q.22.1 shows one to one mapping.
- In this method, the operating system allocates data structure that represents kernel threads. Here multiple threads are run in parallel on multiprocessor system.
- Number of threads in the system increases, the amount of memory required is also increases.
- Windows 95/XP and Linux operating system uses this one to one thread mapping.

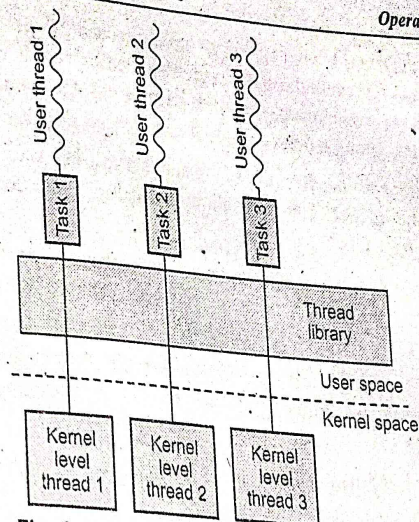


Fig. Q.22.1 One to one multithreading model

- Only overhead in this method is creation of kernel level thread for user level thread. Because of this overhead, system performance is slowdown.

2. Many to one

- Many to one mapping means many user thread maps with one kernel thread. Fig. Q.22.2 shows many to one mapping.

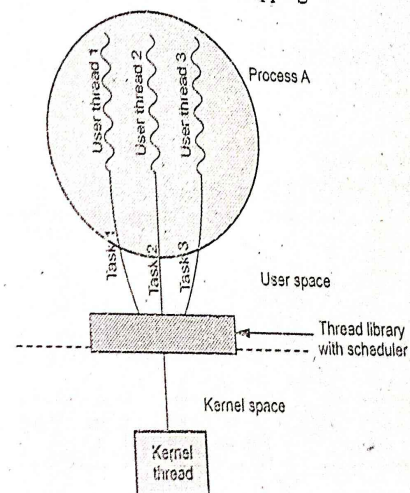


Fig. Q.22.2 Many to one multithreading model

- Operating system blocks the entire multi-threaded process when a single thread blocks because the entire multi-threaded process is a single thread of control. So when operating system receive any a blocking I/O request, it blocks the entire process.
- Thread library handle thread management in user space. The many-to-one model does not allow individual processes to be split across multiple CPUs.
- This type of relationship provides an effective context-switching environment, easily implementable even on simple kernels with no thread support.
- Green threads for Solaris and GNU portable threads implement the many-to-one model in the past, but few systems continue to do so today.
- **Advantage :** System performance is improved by customizing the thread library scheduling algorithm.

3. Many to many

- In the many to many mapping, many user-level threads maps with equal number of kernel threads. Fig. Q.22.3 shows many to many thread mapping.

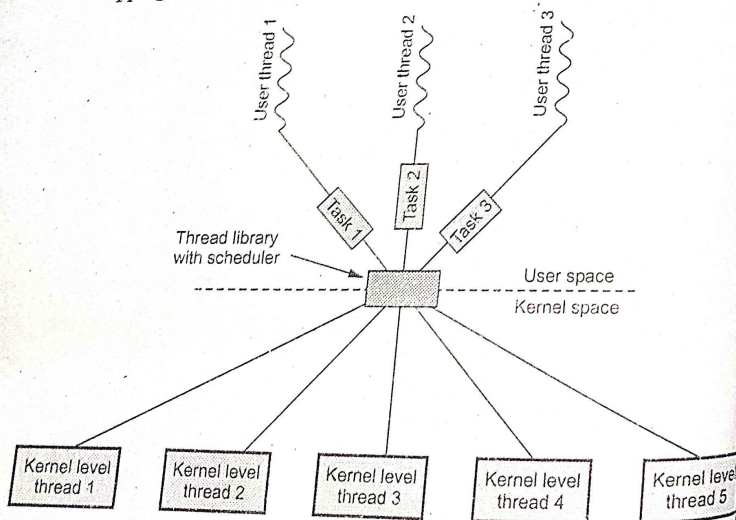


Fig. Q.22.3 Many to many multithreading model

- Many to many mapping is also called M-to-N thread mapping. Thread pooling is used to implement this method.
- Users can create required number of thread and there is no limitation for user. Again here blocking Kernel system calls do not block the entire process.
- This model support for splitting processes across multiple processors.
- **Limitations :** Operating system design becomes complicated.

4.7 : Process Control System Calls

Q.23 How process are created by using system call ? Explain.

Ans. : Operating system creates the process in following situations :

1. Starting of new batch job.
2. User request for creating new process.
3. To provide new services by OS.
4. System call from currently running process.

- Operating system creates a new process with the specified or default attributes and identifier. A process may create several new sub-process.
- Parent process is creating process and the new processes are called the **children** of the process. When operating system creates process, it builds the data structure for managing process and allocates address space in primary main memory.
- Operating system creates foreground and background process. Process is identified by unique Process Identifier (PID) in UNIX and windows operating system. PID value is an integer number.
- All processes in UNIX are created using the `fork ( )` system call. The forking process is called the parent process. The new process is called the **child** process.
- Both the parent and child process have their own and private memory. Open files are shared between parent and child.
- If the parent changes the value of its variable, the modification will only affect the variable in the parent process's address space. Other address spaces created by `fork ( )` calls will not be affected even though they have identical variable names.

- When a process is created, OS assigns some attributes. These are priority, privilege level, requirement of memory, access right, memory protection, PID etc. To perform operation, process needs software and hardware resources. It includes CPU time, files, memory, I/O device.

- Relation between parent process and child process is as follows :

1. Parent process continues to execute concurrently with its child process.
2. Parent process waits until some or all of its children have terminated.

```
void main( )
```

```
{
    printf("Operating System\n");
    fork( );
    printf("Technical Publications\n");
    return 0;
}
```

- In above program Operating System is printed only once and Technical Publications is printed two times.

4.8 : Process Scheduling

Q.24 Explain scheduling queue, ready queue and device queue with respect to process scheduling.

Ans. : • CPU utilization is maximizing by using multiprogramming concept. Processor is not idle, it is executing a process. Processor scheduler selects one process for execution from the ready queue.

Scheduling queue : Scheduling queue is queue of processes or input-output devices. When the user process enters into the system, they put into the job queue. Job queue consists of all processes of the system.

- Operating system maintains different types of queues for different purposes. The queue are ready queue, device queue etc. Fig. Q.24.1 shows ready queue and device queue.
- **Ready queue :** The processes which are ready and waiting to execute are kept in the ready queue. Ready queue is stored in main memory. Linked list is used for representing ready queue. Pointer field of PCB is used for this.

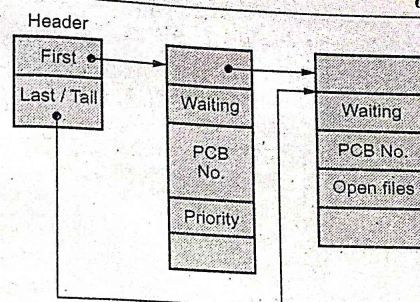


Fig. Q.24.1 Ready and device queue

- **Device queue :** Device queue is a queue for which multiple processes are waiting for a particular I/O device. Each device has maintained its own device queue.
- Process scheduling is represented by queueing diagram. Newly created process is kept into the ready queue and waits for processor. When CPU select the process for executing, following things happens :
 1. Process may require I/O device to perform operation. So it is put into the I/O queue.
 2. Process may create child process and wait for child process termination.
 3. Because of interrupt, process preempted from CPU and put into the ready queue.

4.9 : Scheduling

Q.25 What are the types of schedulers ? Explain them with suitable diagram.

[SPPU : May-19, End Sem, Marks 8]

OR Explain following types of schedulers :

- i) Short term ii) Long term iii) Medium Term.

[SPPU : May-18, End Sem, Marks 6]

Ans. : Schedulers are used to handles process scheduling. It is one type of system software and selects the jobs from the system and decide which process to run. Schedulers are of three types -

1. Long term scheduler
2. Short term scheduler
3. Medium term scheduler

Fig. Q.25.1 shows queueing diagram for process scheduling.

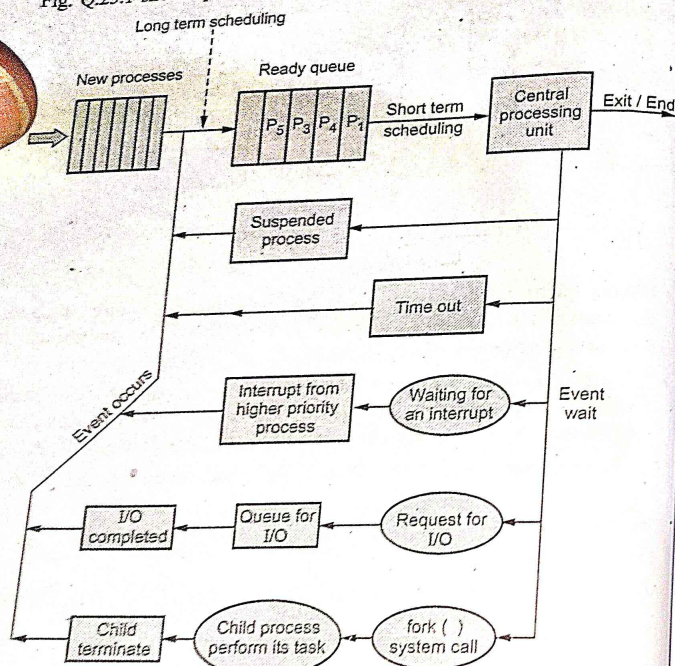


Fig. Q.25.1 Process scheduling queueing diagram

Long term scheduler

- Long term scheduler is also called job scheduler. It determines which process are admitted to the system for processing. Processes are selected from the queue and loads into the main memory for execution.
- Long term scheduling controls the degree of multiprogramming in multitasking systems. It provides a balanced mix of jobs, such as I/O bound and CPU bound.
- Long term scheduling is used in real time operating system. Time sharing operating system has no long term scheduler.

Medium term scheduler

- Medium term scheduler is part of swapping function. Sometimes it removes the process from memory. It also reduces the degree of multiprogramming.
- If process makes an I/O request and it is in memory then operating system takes this process into suspended states. Once the process becomes suspended, it cannot make any progress towards completion.
- In this situation, the process is removed from memory and makes free space for other process.
- The suspended process is stored in the secondary storage device i.e. hard disk. This process is called swapping.

Short term scheduler

- Short term scheduler is also called CPU scheduler. It selects the process from queue which are ready to execute and allocate the CPU for execution.
- Short term scheduler is faster than long term scheduler. This scheduler makes scheduling decisions much more frequently than the long-term or mid-term schedulers.
- A scheduling decision will at a minimum have to be made after every time slice, and these are very short.
- It is also known as dispatcher.

Q.26 Discuss the difference among short term, medium term and long term scheduling.

Ans. :

Sr. No.	Long term	Short term	Medium term
1.	It is job scheduler.	It is CPU scheduler.	It is swapping.
2.	Speed is less than short term scheduler.	Speed is very fast.	Speed is in between both.
3.	It controls the degree of multiprogramming.	Less control over degree of multiprogramming.	Reduce the degree of multiprogramming.
4.	Absent or minimal in time sharing system.	Minimal in time sharing system.	Time sharing system use medium term scheduler.

5.	It select processes from pool and load them into memory for execution.	It select from among the processes that are ready to execute.	Process can be reintroduced into memory and its execution can be continued.
6.	Process state is (New to Ready).	Process state is (Ready to Running)	-
7.	Select a good process, mix of I/O bound and CPU bound.	Select a new process for a CPU quite frequently.	-

Q.27 Define and explain context switching.

Ans. : • A context switch is the switching of the CPU from one process or thread to another. A context is the contents of a CPU's registers and program counter at any point in time.

- Switching the CPU to another process requires performing a state save of the current process and a state restore of a different process. This task is known as a context switch.
- A context switch can mean a register context switch, a task context switch, a thread context switch or a process context switch.
- A register is a small amount of very fast memory inside of a CPU that is used to speed the execution of computer programs by providing quick access to commonly used values.
- A program counter is a specialized register that indicates the position of the CPU in its instruction sequence and which holds either the address of the instruction being executed or the address of the next instruction to be executed, depending on the specific system.
- Context switching can be described in more detail as the Kernel performing the following activities with regard to processes (including threads) on the CPU :
 1. Suspending the progression of one process and storing the CPU's state (i.e., the context) for that process somewhere in memory.
 2. Retrieving the context of the next process from memory and restoring it in the CPU's registers and
 3. Returning to the location indicated by the program counter in order to resume the process.

- Context switches can occur only in Kernel mode (system mode). Kernel mode is a privileged mode of the CPU in which only the Kernel runs and which provides access to all memory locations and all other system resources.
- Other programs, including applications, initially operate in user mode, but they can run portions of the Kernel code via system calls. Software context switching can be used on all CPUs and can be used to save and reload only the state that needs to be changed.
- To use the hardware context switch you need to tell the CPU where to save the existing CPU state and where to load the new CPU state from. The CPU state is always stored in a special data structure called a TSS (Task State Segment).
- Context switch times are highly dependent on hardware support. Context switching represents a substantial cost to the system in terms of CPU time and it can be the most costly operation on an operating system.
- There are three situations where a context switch needs to occur. They are multitasking, interrupt handling, user and Kernel mode switching.

4.10 : Scheduling Algorithms :
FCFS, SJF, RR and Priority

Q.28 Explain preemptive and non-preemptive scheduling.

Ans. : • The scheduling policy determines when it is time for process to be removed from the CPU and which ready process should be allocated the CPU next. The scheduling mechanism is composed of several different parts, depending on exactly how it is implemented in any particular operating system.

- CPU scheduling is divided into two types : Preemptive scheduling and non-preemptive scheduling.
- **Preemptive scheduling :** A scheduling method that interrupts the processing of a process and transfers the CPU to another process is called a preemptive CPU scheduling. The process switches from running state to the ready state and waiting state to the ready state.

- **Non-preemptive scheduling** : Non-preemptive operation usually proceeds towards completion uninterrupted. Once the system has assigned a processor to a process, the system cannot remove that processor from the process. The process switches from running state to the waiting state and termination of process.
- Preemptive scheduling increases the cost and it has higher overheads. This scheduling method is useful only in the high priority processes require rapid response.
- Non-preemptive scheduling is simple to implement. It requires some hardware platform. This method is attractive because of its simplicity. Windows 3.1 and Apple Macintosh operating system uses this scheduling method.

Q.29 Explain difference between preemptive and Non-preemptive scheduling.

Ans. :

Sr. No.	Preemptive scheduling	Non-preemptive scheduling
1.	Preemptive scheduling allows a process to be interrupted in the midst of its execution, taking the CPU away and allocating it to another process.	Non-preemptive scheduling ensures that a process relinquishes control of the CPU only when it finishes with its current CPU burst.
2.	Preemptive scheduling incurs a cost associated with access shared data.	Non-preemptive scheduling does not increase the cost.
3.	It also affects the design of the operating system Kernel.	It does not affect the design of OS Kernel.
4.	Preemptive scheduling is more complex.	Simple, but very inefficient.
5.	Example : Round robin method.	Example : First come first serve method.

Q.30 Explain CPU scheduling criteria.

- Ans. : • Depending on the system, CPU scheduling criteria will change.
1. **Throughput** : CPU scheduling should attempt to service the maximum number of processes per unit time. The higher is the number, the more work is done by the system.
 2. **Waiting time** : The average period of time a process spends waiting. Process is normally in the ready queue in waiting time.
 3. **Turnaround time** : Turnaround time start from process submission to completion of process.

$$\text{Turnaround time} = \text{Burst time} + \text{Waiting time}$$

4. **Response time** : It is the time from the submission of a request until the first response is produced.
 5. **CPU utilization** : It is average function of time during which the processor is busy.
 6. **Fairness** : Avoid the process from the starvation. All the processes must be given equal opportunity to execute.
 7. **Priority** : If the operating system assigns priorities to processes, the scheduling mechanism should favor the higher priority processes.
 8. **Predictability** : A given process always should run in about the same amount of time under similar system loads.
- Depending upon the nature of operations the scheduling policy may differ. The CPU utilization and throughput are the system centered parameters. Fairness is affect by user and system.

Q.31 What is CPU scheduling ? Explain 2 different scheduling algorithms with examples.

Ans. : **CPU scheduling** : • In a multiprogramming environment, usually more programs to be executed than could possibly be run at one time. In CPU scheduling, it switches from one process to another process. CPU resource management is commonly known as scheduling.

- Objective of the multiprogramming is to increases the CPU utilization. CPU scheduling is one kind of fundamental operating system functions.
- User program contains combination of CPU burst cycle and I/O burst cycles. Process starts with CPU burst cycle then I/O burst cycle again

CPU burst cycle again I/O burst cycle. In this way the process completes its executions. Process will terminate after final CPU burst cycle completions.

- Scheduling mechanism is the part of the process manager that handles the removal of the running process from the CPU and the selection of another process on the basis of a particular strategy.
- Scheduler is responsible for multiplexing processes on the CPU.

1. First In First Out (FIFO) scheduling : • FIFO is also called first come first served scheduling method. The processes are dispatched according to their arrival time at the ready queue. FIFO is non-preemptive CPU scheduling algorithm.

- FIFO is simple to implement because it uses a FIFO queue. This algorithm is fine for most of the batch operating systems. FIFO is not useful in scheduling interactive processes because it cannot guarantee short response time.

- When new process enters into the system, its process control block is linked to the end of the ready queue and it is removed from the front of the queue.

- Turnaround time is unpredictable with the FIFO algorithm. Average waiting time of the FCFS is often quite long. Real life analogy is buying tickets.

- Convey effect : To reduce I/O device utilization, all I/O bound processes will be waiting excessively long for processor bound ones. This is called as convey effect.

- If one process monopolizes the system, the extent of its overall effect of system performance depends on the scheduling policy and whether the process is processor bound or I/O bound.

2. Priority scheduling : • Priority scheduling is preemptive or non-preemptive. Priority of the process can be defined either internally or externally. Internally defined priority considers the time limits, number of open files, use of memory and use of I/O devices.

- External priorities are set by using external parameter of the process, like importance of a process, cost of process etc.
- In this scheduling algorithm, CPU select higher priority process first. If the priority of two process is same then FCFS scheduling algorithm is applied for solving the problem.

- A non-preemptive priority algorithm will simply put the new process at the head of the ready queue. In non-preemptive scheduling algorithm, currently executing process will not change state.
- A preemptive priority scheduling algorithm will preempt the CPU if the priority of the newly arrived process is higher than the priority of the currently running process.

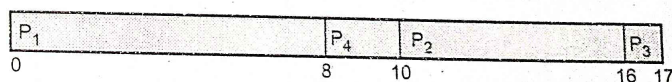
Drawback of priority scheduling :

- Waiting time is more for lower priority process even if there required CPU burst time is less. Priority scheduling algorithm faces the starvation problem.
- Starvation problem can be solved by using aging techniques.
- In aging, the priority of the process will increase which is waiting for long time in the ready queue.

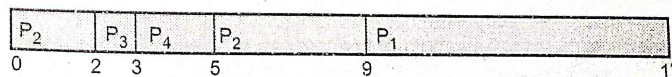
Q.32 Draw Gantt chart and calculate average turnaround time, average waiting time for the following processes using priority based (no-preemptive) scheduling and SJF (preemptive) scheduling policies. (Consider low no. as high priority) [SPPU : May-19, End Sem, Marks 5]

Processes	Arrival time	Burst time	Priority
P ₁	0	8	1
P ₂	0	6	2
P ₃	2	1	3
P ₄	3	2	0

Ans. : Priority based (Non-preemptive)



SJF preemptive :



Process	Waiting time		Turnaround time	
	Priority non-preemptive	SJF preemptive	Priority non-preemptive	SJF preemptive
P ₁	0	9	8	17
P ₂	10	3	16	9
P ₃	14	0	15	1
P ₄	5	0	7	2

Average waiting time priority non-preemptive = $(0 + 10 + 14 + 5) / 4$
 $= 7.25$

Average waiting time SJF preemptive = $(9 + 3 + 0 + 0) / 4 = 3$

Average turnaround time priority non-preemptive = $(8 + 16 + 15 + 7) / 4$
 $= 11.5$

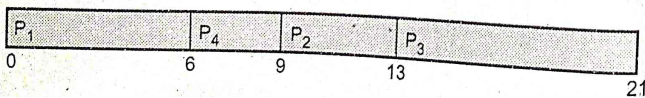
Average turnaround time SJF preemptive = $(17 + 9 + 1 + 2) / 4 = 7.25$

Q.33 Draw Gantt chart and calculate average turnaround time, average waiting time for the following processes using SJF non-preemptive and round robin with time quantum 2.

[SPPU : Dec.-18, End Sem, Marks 6]

Process	Arrival time	Burst time
P ₁	0	6
P ₂	1	4
P ₃	4	8
P ₄	3	3

Ans. : SJF Non-preemptive : Gantt chart :



RR (Time quantum = 2)



Process	Waiting time		Process	Turnaround time	
	SJF Non-preemptive	RR		SJF Non-preemptive	RR
P ₁	0	8	P ₁	6	14
P ₂	9 - 1 = 8	7	P ₂	12	11
P ₃	13 - 4 = 9	9	P ₃	17	17
P ₄	6 - 3 = 3	9	P ₄	6	12

Average waiting time for SJF non-preemptive = $(0 + 8 + 9 + 3) / 4$
 $= 20 / 4 = 5$

Average waiting time for RR = $(8 + 7 + 9 + 9) / 4 = 33 / 4 = 8.25$

Average turnaround time for SJF non-preemptive = $(6 + 12 + 17 + 6) / 4$
 $= 41 / 4 = 10.25$

Average turnaround time for RR = $(14 + 11 + 17 + 12) / 4$
 $= 54 / 4 = 13.5$

Q.34 What is pre-emptive scheduling ? Consider following example and supply round robin scheduling policy and calculate average Turnaround (TT) Time. (consider time slice = 1).

[SPPU : Dec.-19, End Sem, Marks 6]

Process ID	Arrival time	Serve time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

Ans. : Gantt chart for Round Robin (Time slice = 1)

A	A	B	A	B	C	B	C	D	B	C	E	D	B	C	E	D	B	D	D
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

Process ID	Waiting time	Turnaround time
A	1	4
B	10	16
C	7	11
D	9	14
E	6	8

$$\text{Average turnaround time} = \frac{4+16+11+14+8}{5} = 10.6$$

END...